Welcome

Digital and Computer Vision

Lesson 1: Session 1

Introduction to Computer Vision

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Introduction of the Professor

I am from Bulgaria, but currently also have Switzerland nationality.

I am a professor of AI and Computer Vision.

I have a PhD in physical chemistry from Bulgarian Academy of Science. During my PhD, I studied fractal aggregates of nanoparticles and developed several programs for Monte Carlo computer simulations. This experience sparked my passion for computer science and programming.

Introduction of the Professor

- I have eight years of teaching experience at UniGe and EPFL, working with both bachelor's and master's students. Additionally, I have developed algorithms and code for several startup companies in the fields of Biotech, Computer Vision, and Fintech.

- Between 2000 and 2003, I was part of a startup called Manteia Predictive Medicine, where we developed a high-throughput DNA sequencing method based on the analysis of microscopy images. The company was acquired in 2003 by the British firm Solexa, and today, this method remains the primary technology used for DNA sequencing.

- Currently I am professor at SWISS UMEF

Course Presentation

Lesson 1: Introduction to Computer Vision and Digital Image Processing.

Lesson 2: Digital Image Processing : smoothing, thresholding, edge and contours detection.

Lesson 3: Computer Vision Techniques : segmentation, feature and object detection, morphological operations.

Lesson 4: Deep Learning for Computer Vision : CNNs, TensorFlow, Face Recognition

Lesson 5: Final Project and Course Recap

Summary of the current session: Introduction to Computer Vision

In this session we will covered the following

What is Computer Vision? Applications in AI, robotics, and real-world industries.
Understanding digital image properties, formats, channels, size, pixels, RGB, Grayscale).
Overview of OpenCV. Installing OpenCV and setting up the Python environment.
Exercise: Reading, displaying, and saving images using OpenCV.

Course Objectives&value

- Understand the fundamentals of Computer Vision and his applications.
- Develop and lead real-world Computer Vision projects.

What is Computer Vision?

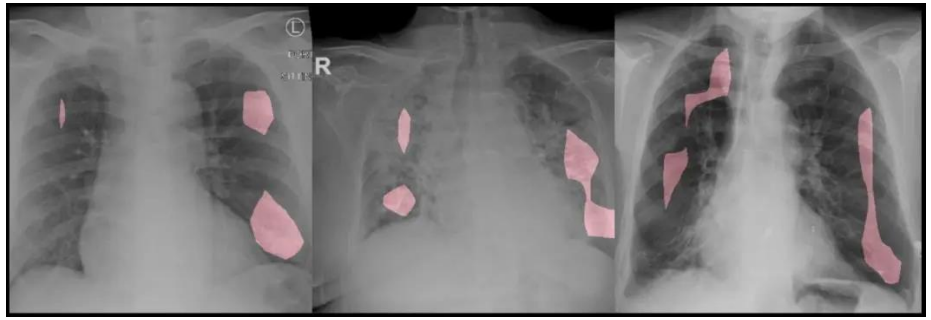
- Computer Vision is a field of Artificial Intelligence (AI) that enables machines to see, interpret, and understand visual information from the world, simulating the way humans do.

How Does Computer Vision Work?

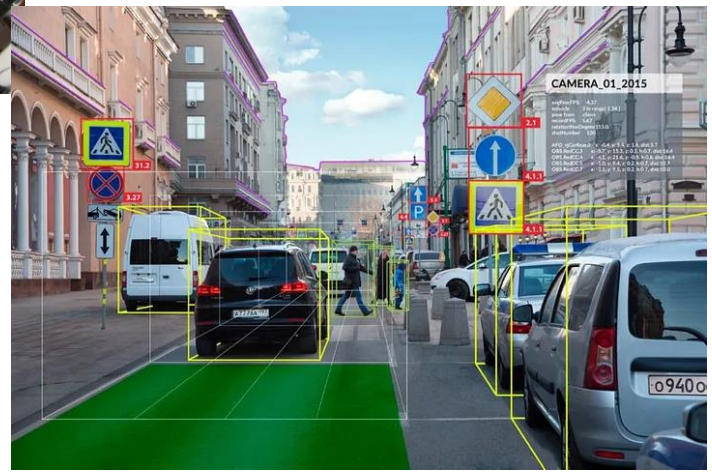
- Computer Vision relies on image processing techniques, machine learning, and deep learning to analyze and understand visual data. The basic steps include:

- 1. Image Acquisition 📷 ⟶ Capture images or video from cameras, sensors, or files.
- 2. Preprocessing 🔍 ⟶ Convert to grayscale, resize, normalize, or filter noise.
- 3. Feature Extraction 📊 ⟶ Detect edges, objects, faces, or patterns.
- 4. Object Recognition 🎯 ⟶ Use AI models to classify and recognize objects.
- 5. Decision Making 🧠 ⟶ Use AI to take actions based on visual data (e.g., detect a pedestrian and stop a car).

Applications of Computer Vision



https://viso.ai/applications/computer-vision-in-healthcare/



https://becominghuman.ai/computer-vision-applications-in-self-driving-cars-610561e14118

Digital Image

A digital image is essentially a two-dimensional matrix (or three-dimensional in the case of colored images), defined by the mathematical function f(x, y).

1. Pixel Values: An image is made up of pixels, which are the smallest units of information in an image. Each pixel has a value that represents its color and intensity.

2. Digital Representation: The RGB values are usually represented as integers ranging from 0 to 255. 0 represents the absence of color (black), and 255 represents the maximum intensity of that color (full brightness).

3. Image Matrix: The computer reads the image as a matrix of numbers, where each element in the matrix corresponds to the pixel value at that location. For a color image, there are typically three matrices, one for each RGB channel.

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & f(0,2) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & f(1,2) & \cdots & f(1,N-1) \\ \vdots & \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & f(M-1,2) & \cdots & f(M-1,N-1) \end{bmatrix}$$

Color Spaces

- Color spaces are a way to represent the color in the image. There are several different color spaces and each has its own significance. Some of the popular color spaces are RGB (Red, Green, Blue), CMYK (Cyan, Magenta, Yellow, Black), HSV (Hue, Saturation, Value), etc. OpenCV's default color space is RGB. However, it actually stores color in the BGR format.
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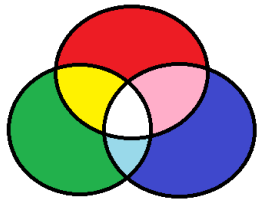
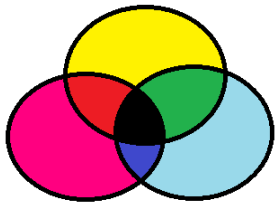
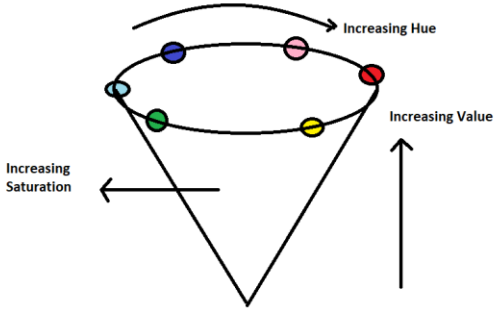
- https://www.geeksforgeeks.org/color-spaces-in-opencv-python/



RGB color space

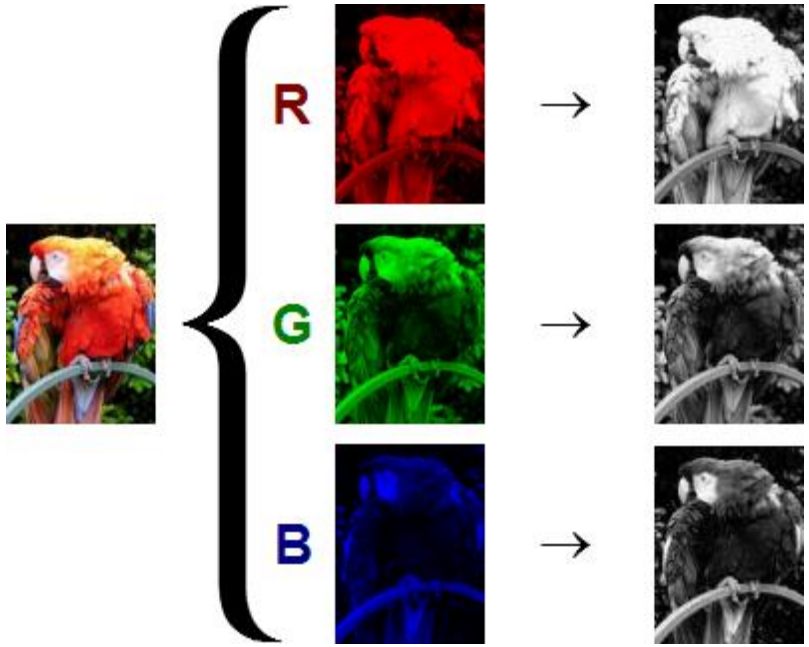


CMYK color space



HSV color space

Channels



- This work has been released into the public domain by its author, Ricardo Cancho Niemietz at English Wikipedia. This applies worldwide.

Formats

Raster Images – Made of pixels, resolution-dependent (JPEG, PNG, BMP).

Vector Images – Made of mathematical equations, scalable (SVG, EPS).

Format	Best Use Case	Pros	Cons
JPEG	Photography, web images	Small size, good quality	Compression, loses some quality
PNG	Logos, transparency	High quality, lossless	Larger file size
GIF	Animations, memes	Supports animation	256 colors only
BMP	Image processing	High quality	Very large size
TIFF	Printing, professional work	Best quality	Huge file size
WEBP	Modern web images	Small + high quality	Limited browser support
RAW	Professional photography	Highest detail	Requires processing

What is OpenCV Library?

OpenCV, short for Open Source Computer Vision, is an open-source computer vision and machine learning software library. Originally developed by Intel, it is now maintained by a community of developers under the OpenCV Foundation. By using it, one can process images and videos to identify objects, faces, or even the handwriting of a human. The first OpenCV version was 1.0. OpenCV is released under a BSD license and hence it's free for both academic and commercial use. It has C++, C, Python, and Java interfaces and supports Windows, Linux, Mac OS, iOS and Android. When opencv was designed the main focus was real-time applications for computational efficiency. All things are written in optimized C/C++ to take advantage of multi-core processing.

https://www.geeksforgeeks.org/opencv-overview/

Why Python?

OpenCV provides a fully functional Python API, which allows access to all core functionalities of the library.

Python have many libraries (NumPy, Matplotlib, TensorFlow and PyTorch) and allows fast development and testing of algorithms, which is crucial for research and innovation in computer vision and AI.

OpenCV with Python works on Windows, Linux, macOS, and even embedded systems, making it versatile for different environments.

Python and OpenCV have large, active communities, which means extensive documentation, tutorials, and forums for problem-solving.

What is NumPy?

NumPy (Numerical Python) is a powerful Python library for numerical computing. It provides multi-dimensional arrays (ndarrays) and a collection of mathematical functions to operate on them efficiently.

Images in OpenCV are NumPy arrays: OpenCV represents images as NumPy arrays, making it easy to manipulate pixel values.

Fast operations: NumPy enables efficient image processing (e.g., filtering, transformations).

Seamless integration: Many OpenCV functions work directly with NumPy arrays.

It is not necessary to explicitly install NumPy when installing OpenCV, as it will be automatically included as a dependency.

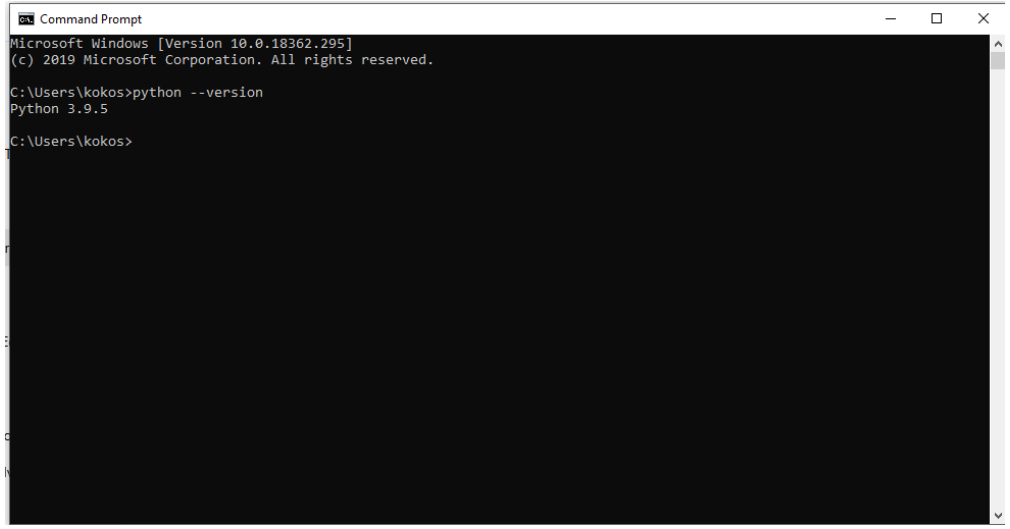
Installing OpenCV and the Python environment

Do you have Python?

Open the Command line prompt as **administrator** (search for cmd in the Run dialog(+ R). Now run the following command:

python --version

If not go to: https://www.geeksforgeeks.org/how-to-install-python-on-windows/



Installing Python on Windows - 1

1. Visit the official page for Python: https://www.python.org/downloads/

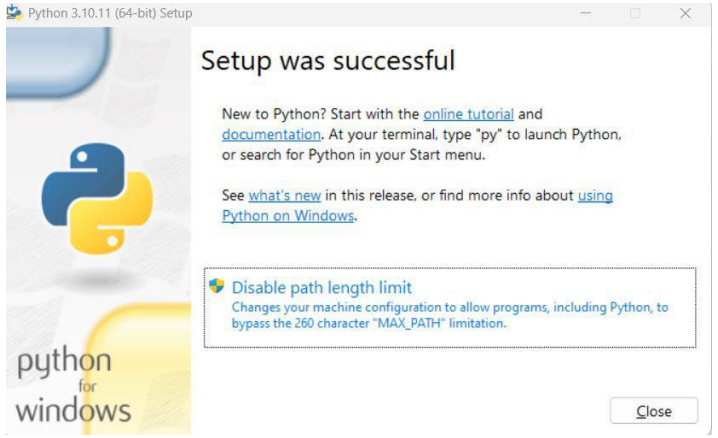
Locate a reliable version of Python 3, preferably version 3.10.11, which was used in testing this tutorial. Choose the correct link for your device from the options provided: either Windows installer (64-bit) or Windows installer (32-bit) and proceed to download the executable file.

2. Once you have downloaded the installer, open the .exe file, such as python-3.10.11-amd64.exe, by double-clicking it to launch the Python installer. Choose the option to Install the launcher for all users by checking the corresponding checkbox, so that all users of the computer can access the Python launcher application. **Enable users to run Python from the command line by checking the Add python.exe to PATH checkbox.**

Clic on the **Install Now** Button

Installing Python on Windows - 2

3. After completing the setup. Python will be installed on your Windows system. You will see a successful message.



4. Close the window after successful installation of Python. You can check if the installation of Python was successful by using either the command line or the Integrated Development Environment (IDLE), which you may have installed. To access the command line, click on the Start menu and type “cmd” in the search bar. Then click on Command Prompt.

Restart your computer and check again:

python --version

Visual Studio Code

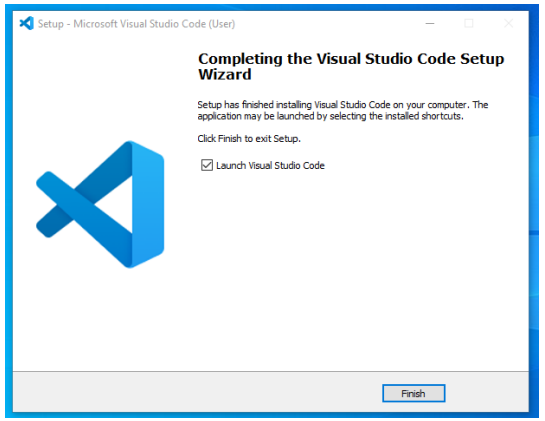
Visual Studio Code (VS Code) is a free, lightweight, and open-source code editor developed by Microsoft. It supports multiple operating systems, including Windows, macOS, and Linux.

VS Code is designed to be versatile and user-friendly, offering features such as:

Syntax Highlighting and Autocompletion: Enhances code readability and writing efficiency.
Integrated Debugging: Allows developers to test and debug code within the editor.
Built-in Git Integration: Facilitates version control and collaboration directly from the editor.
Extensibility: Supports a wide range of extensions to add functionalities like new language support, themes, and tools.

Steps to Install Visual Studio Code on Windows

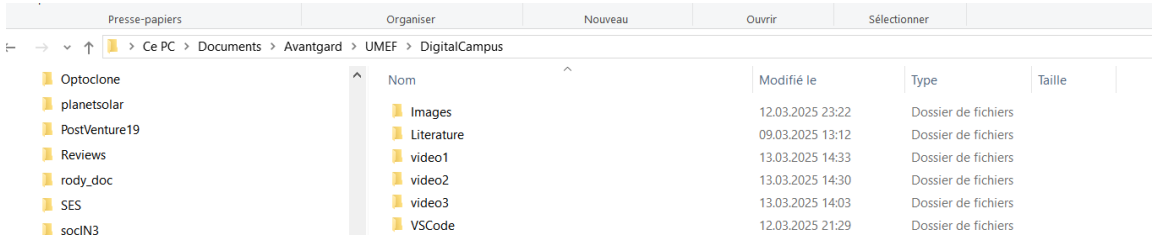
1. Download the Visual Studio Code on the Official Website: https://code.visualstudio.com/docs/?dv=win
2. Click on the Installer icon to start the installation process of the Visual Studio Code.
3. After the Installer opens, it will ask you to accept the terms and conditions of the Visual Studio Code. Click on I accept the agreement and then click the Next button.
4. Choose the location data for running the Visual Studio Code. It will then ask you to browse the location. Then click on the Next button.
5. Then it will ask to begin the installation setup. Click on the Install button.
6. After the Installation setup for Visual Studio Code is finished, it will show a window like this below. Click “Finish” Then restart your computer



Set Up your environment

1. Add folders to store Images:
Go to the web on github and download the images which will be used in our CV course :

https://github.com/opencv/opencv/tree/master/samples/data

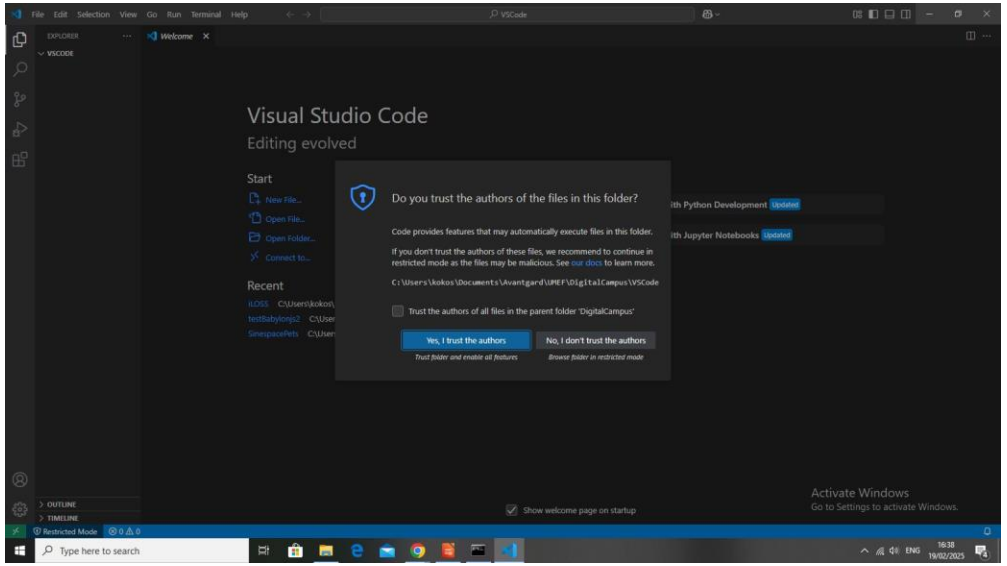


Create a folder to store your OpenCV project. I call this VSCode, but you can put any valid name.

Start your VS Code program, go to:
File → Open Folder

Navigate to your VSCode folder and open it.

Confirm you trust the authors.



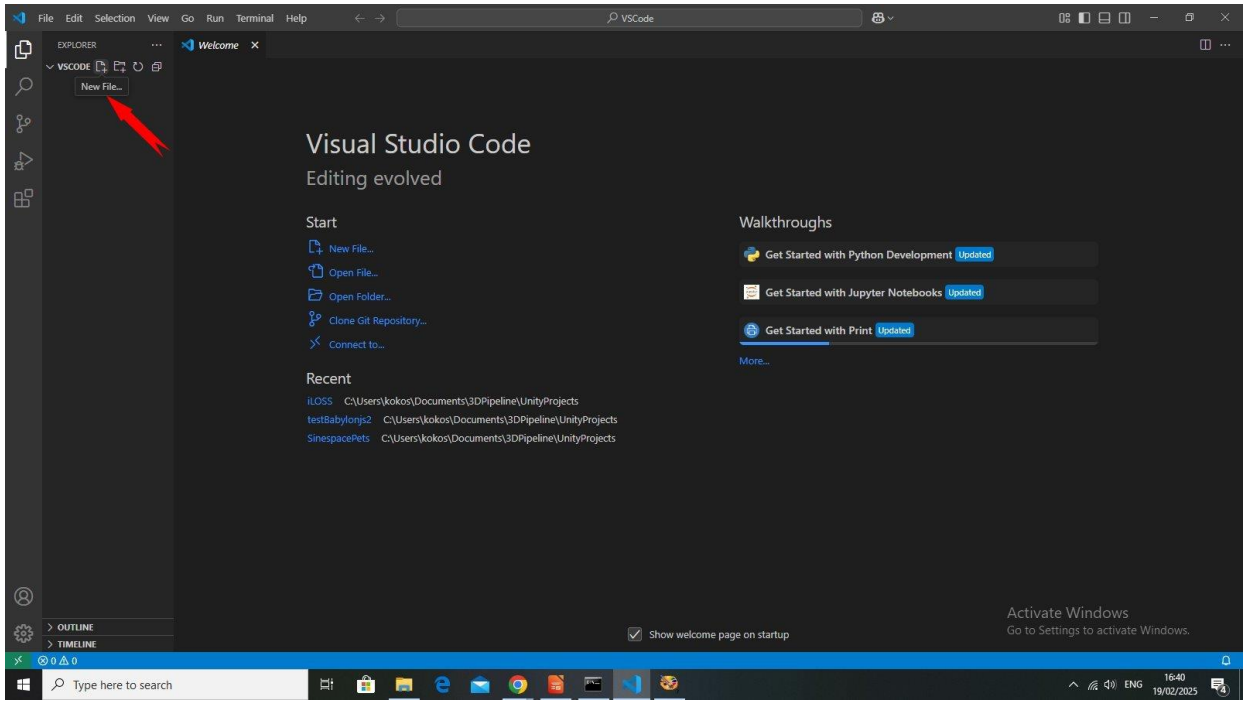
Create a very simple python script

Create a python file for example test.py by click on add new file to the left up corner or File → New File

Type in the editor:

```
import cv2
print(cv2.__version__)
```

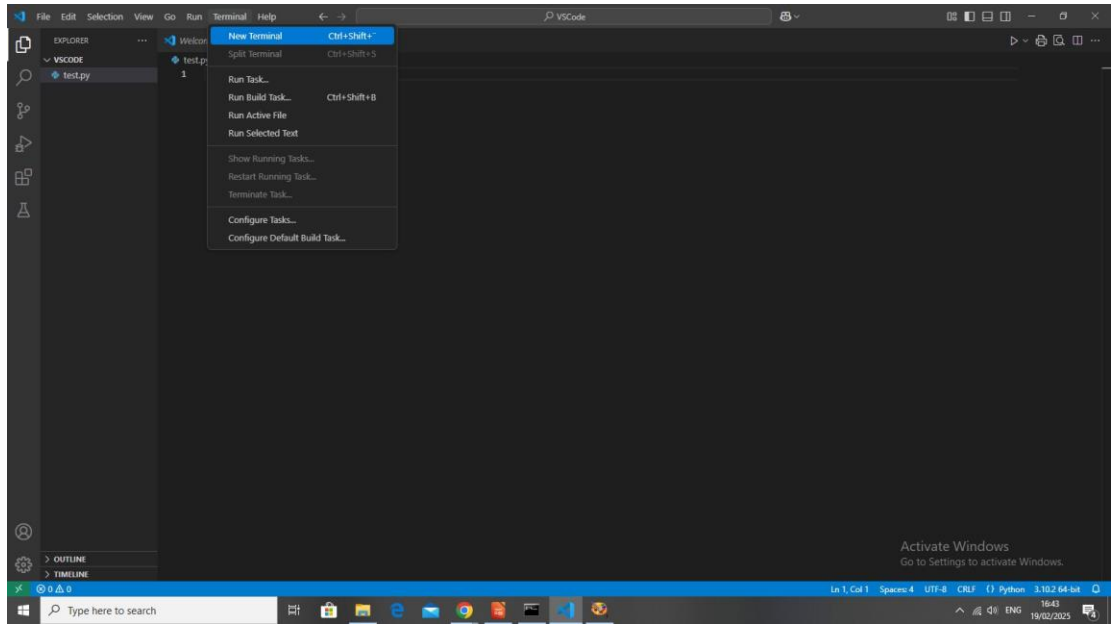
Then save the file:
File->Save



Create new virtual environment in a terminal

Start a terminal by click:
Terminal →New Terminal

Set the Terminal type as command prompt (cmd) and not a Power Shell in the low right corner of the screen.



Type:
python -m venv myenv

Where myenv is the name of the environment.
Respond on the next question with “yes”

Activate the environment by type:

myenv\Scripts\activate
And type Enter

Install OpenCV and run the first OpenCV script

Type:

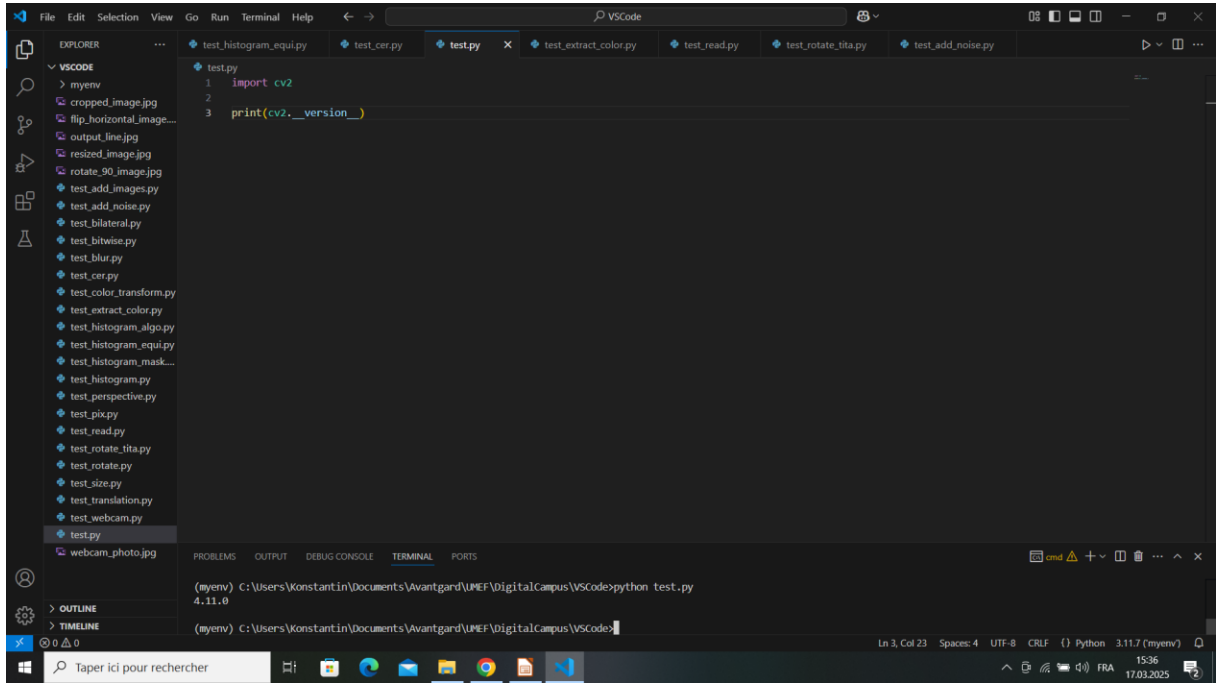
pip install opencv-python

Then Return

Execute the file by typing in the terminal:

python test.py

You will see the version of the OpenCV in the cmd



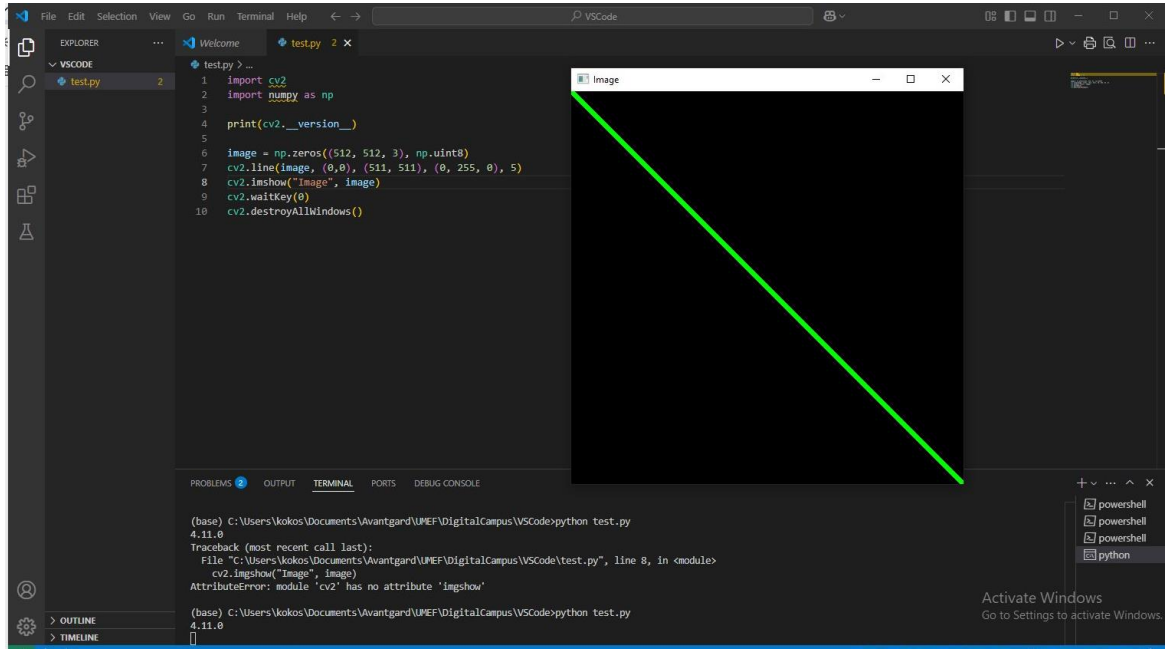
Show a image on OpenCV with python

Type the following script save and run it again:

```
import cv2
import numpy as np

image = np.zeros((512, 512, 3), np.uint8)
cv2.line(image, (0,0), (511, 511), (0, 255, 0), 5)
cv2.imshow("Image", image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

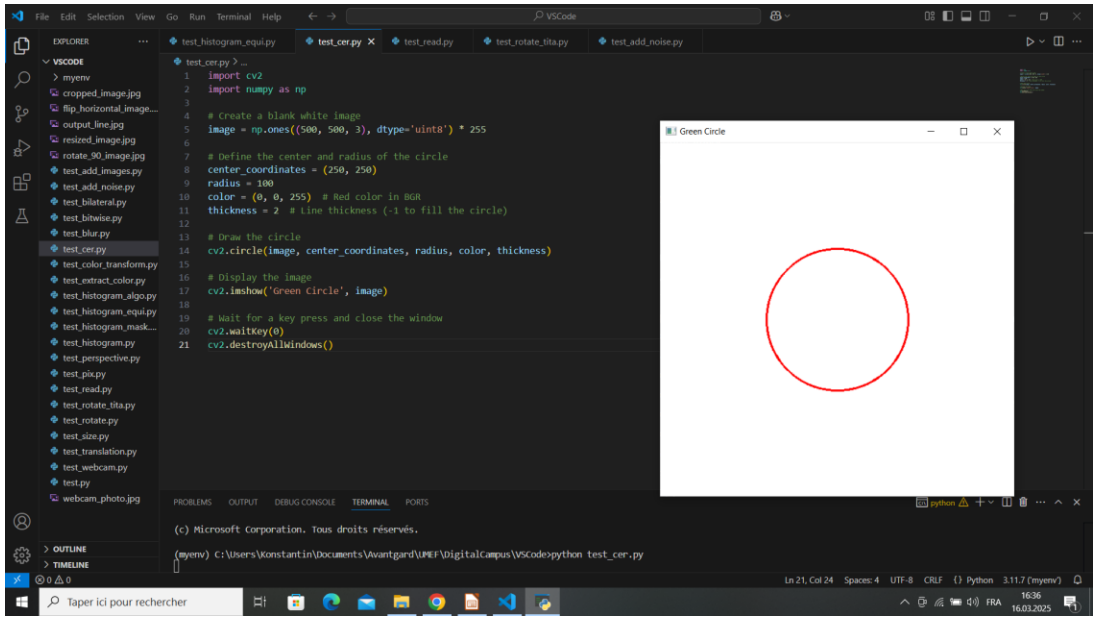
The result is shown left on the screen.
Type any key to close the image



Show a image on OpenCV with python

```
import cv2
import numpy as np

# Create a blank white image
image = np.ones((500, 500, 3), dtype='uint8') * 255
# Define the center and radius of the circle
center_coordinates = (250, 250)
radius = 100
color = (0, 0, 255)  # Red color in BGR
thickness = 2  # Line thickness (-1 to fill the circle)
# Draw the circle
cv2.circle(image, center_coordinates, radius, color, thickness)
# Display the image
cv2.imshow('Green Circle', image)
# Wait for a key press and close the window
cv2.waitKey(0)
cv2.destroyAllWindows()
```



Exercise: Reading, displaying, and saving images using OpenCV

Exercise 1: Save on the disk the constructed image as jpg file using imsave.

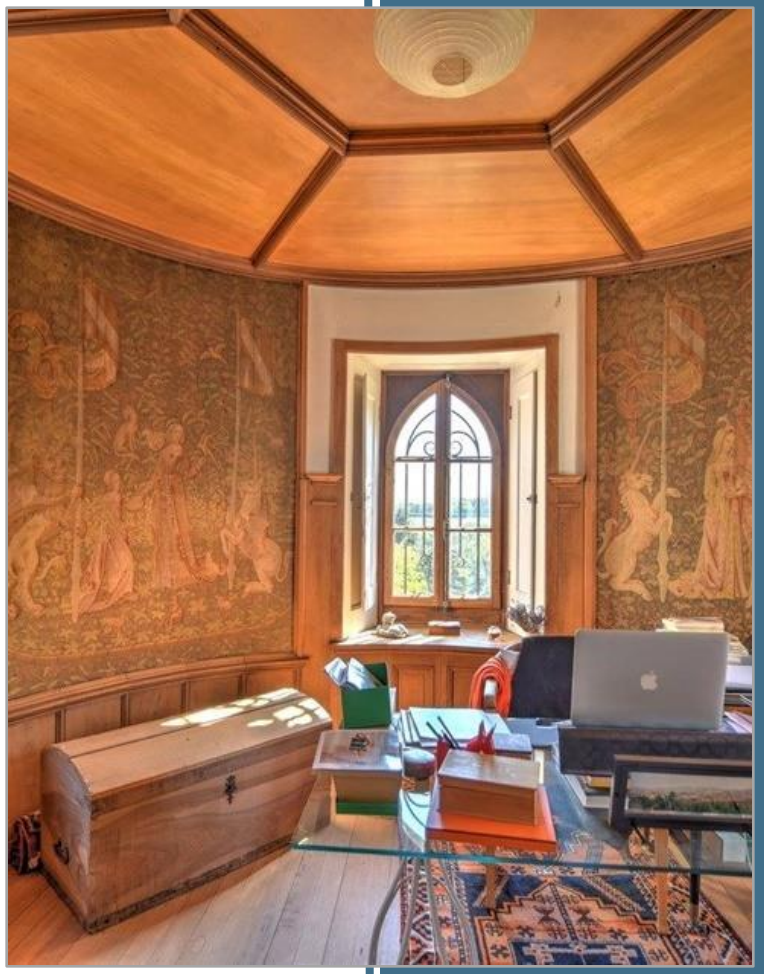
Exercise 2: Read and show this image with imread.

Exercise 3: Read and show another image on your choice from the disk.

In this session we have covered:

What is Computer Vision? Applications in AI, robotics, and real-world industries.
Understanding digital image properties, formats, channels size, shape, pixels, RGB, Grayscale).
Overview of OpenCV. Installing OpenCV and setting up the Python environment.

In the next session we will covered:

Image Basics & Manipulation
Accessing pixel values.
Resizing, cropping, flipping and rotating images.
Scaling, and translating images.
Perspective transformations and warping.
Converting images (Grayscale, HSV, RGB).
Image arithmetic (adding, subtracting, blending images).

THANKS !!!